

Evaluation Sets

Project: Verified Study Buddy

1. Your mission

Your learning target

I can explain the mechanism and terminology of system evaluation sets.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Quality Rubrics. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Block to Python Bridge. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 26: Evaluation Sets

A good eval set represents real use, edge cases, and known risks.



CREATOR MISSION

Author ten tests with expected behavior.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Latency cost and resource use. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Readiness for Ranking and retrieval evaluation

Recall Lexical and vector retrieval. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

My readiness explanation

This week's end goal

By the end, I can explain evaluation sets using a traced example and a stated limitation.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use evaluation sets to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A system evaluation set contains cases with expected behaviour and scoring rules. It should cover retrieval, support, fallback and permissions rather than only easy answerable questions.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Apply the connected idea: Ranking and retrieval evaluation

A ranking needs task-specific relevance judgments and a declared cutoff. Precision and recall have different denominators.

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3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



IN

INPUT

Author ten tests with expected behavior.



SYSTEM

A good eval set represents real use, edge cases, and known risks.



OUT

EVIDENCE

Author ten tests with expected behavior.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Record case ID, evidence version, expected claims or fallback, permitted actions and scoring rubric. Separate development from final cases. Aggregate metrics only after checking required conditions individually.

System evaluation sets — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Ten cases include six answerable and four missing-evidence cases. System answers five answerable correctly and abstains appropriately on three missing cases.

Check or calculation: Record case ID, evidence version, expected claims or fallback, permitted actions and scoring rubric. Separate development from final cases. Aggregate metrics only after checking required conditions individually.

Conclusion and limit: Answer correctness 5/6; appropriate fallback 3/4; combined contract passes 8/10 if each case has one required outcome. Different denominators answer different questions.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Apply the connected idea: Ranking and retrieval evaluation

Precision at k is relevant returned items divided by k when exactly k items are returned. Recall at k is relevant returned items divided by all known relevant eligible items. If the relevant set is unknown, do not invent recall.

Ranking and retrieval evaluation — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Relevant documents are A, C, E. Top three results A, B, C contain two relevant items.

Check or calculation: Precision at k is relevant returned items divided by k when exactly k items are returned.

Conclusion and limit: Precision@3=2/3; recall@3=2/3. Duplicates should not count as distinct retrieved documents.

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Connect the picture and the explanation

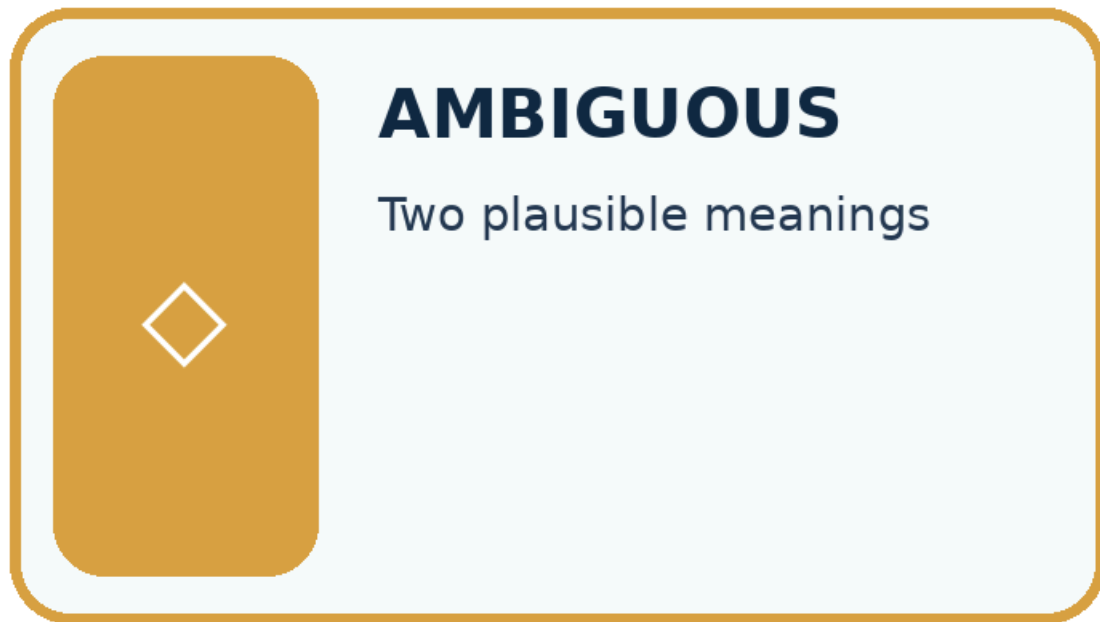
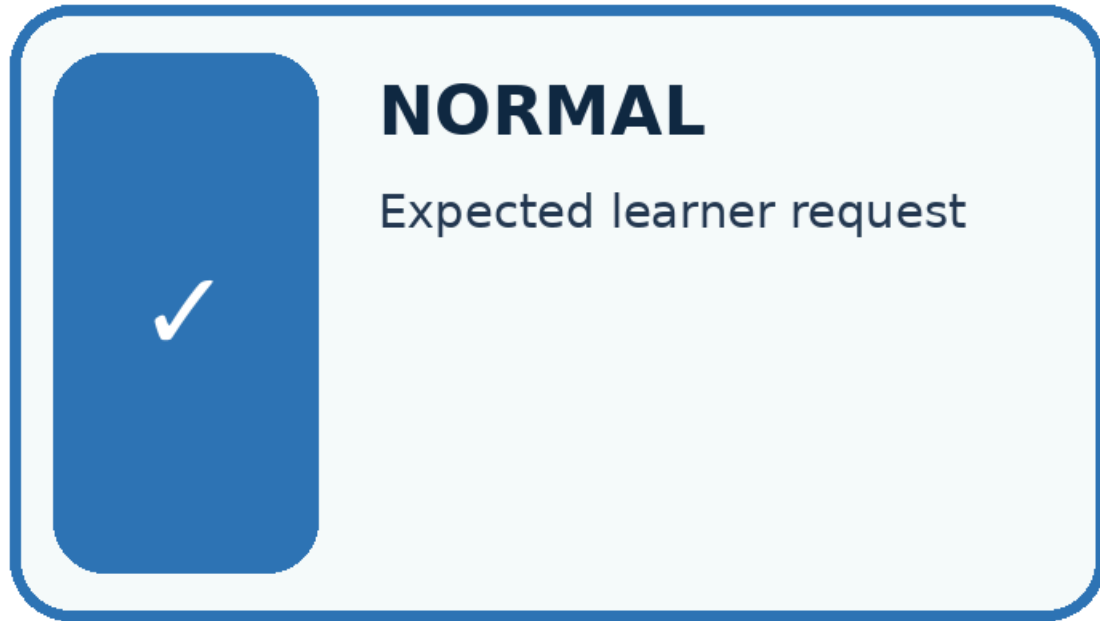
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Add three missing-evidence cases that all pass.

Prediction: Overall becomes 8/9, but supported answering remains 2/3.

Reason: Changing the case mixture raises the pooled rate without improving any existing answerable result.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

COVERAGE

important situations
included

3

EXPECTED BEHAVIOUR

observable acceptable
response

4

REGRESSION TEST

case protecting previously
fixed behavior

Word · My own example · Not this / boundary

Word: evaluation set

My own example: Cases used to measure a system under declared criteria; specify whether they are validation or final test cases.

Not this / boundary: _____

Word: coverage

My own example: The range of relevant cases, conditions or groups included in data or tests.

Not this / boundary: _____

Word: expected behavior

My own example: The result or action required by the specification for a given case, stated before observing the system.

Not this / boundary: _____

Word: regression test

My own example: A repeatable check that previously correct behaviour still works after a change.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Evaluation case

Meaning: An input with evidence and prespecified expected behaviour

Word: Contract pass

Meaning: Satisfying the declared requirements for one case

Terms for Ranking and retrieval evaluation

Term · Meaning

Term: Precision at k

Meaning: Relevant results among the first k returned items

Term: Recall at k

Meaning: Fraction of known relevant eligible items found in the first k results

Term: Cutoff

Meaning: The rank limit used for scoring

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 26 • WORKED EVIDENCE

- 1 Ten cases include factual, comparison, calculation, ambiguous, and missing-source questions.
- 2 Add conflicting-source, prompt-injection, sensitive-data, and accessibility cases.
- 3 Write expected behavior before running the product.
- 4 After a repair, keep the failed case as a regression test.

1. Ten cases include factual, comparison, calculation, ambiguous, and missing-source questions.
2. Add conflicting-source, prompt-injection, sensitive-data, and accessibility cases.
3. Write expected behavior before running the product.
4. After a repair, keep the failed case as a regression test.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Ten cases include six answerable and four missing-evidence cases. System answers five answerable correctly and abstains appropriately on three missing cases.

Answer correctness 5/6; appropriate fallback 3/4; combined contract passes 8/10 if each case has one required outcome. Different denominators answer different questions.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Apply the connected idea: Ranking and retrieval evaluation

Relevant documents are A, C, E. Top three results A, B, C contain two relevant items.

Precision@3=2/3; recall@3=2/3. Duplicates should not count as distinct retrieved documents.

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Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Evaluation Sets

Design an evaluation set that checks answering and appropriate uncertainty separately.

The facts we will use

Six toy questions: three have source answers and three do not. Results: 2 supported answers out of 3; 3 correct “not in source” responses out of 3.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Evaluation Sets

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Separate cases

Keep answerable and missing-evidence cases as different groups.



2. Calculate rates

Answering: 2/3. Appropriate uncertainty: 3/3.



3. Report total

Five of six pass overall; the perfect missing-evidence group hides one answering failure if only totals are discussed.

CHANGE ONE THING

Add three missing-evidence cases that all pass.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Separate cases

What happens and why: Keep answerable and missing-evidence cases as different groups.

Step: 2. Calculate rates

What happens and why: Answering: 2/3. Appropriate uncertainty: 3/3.

Step: 3. Report total

What happens and why: Five of six pass overall; the perfect missing-evidence group hides one answering failure if only totals are discussed.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Five of six pass overall; the perfect missing-evidence group hides one answering failure if only totals are discussed. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

A small constructed set probes specific behaviours; its pooled rate is not an estimate of every future classroom question.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: More random test cases always create better coverage. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

Apply the connected idea: Ranking and retrieval evaluation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

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8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Evaluation coverage limits the claim; prompts tuned on these cases need new reserved final cases.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Apply the connected idea: Ranking and retrieval evaluation

Known judgments may be incomplete; a score over one query does not establish coverage over all queries.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “System evaluation sets”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for System evaluation sets. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Ranking and retrieval evaluation

Use its named workbook W2. Identify the given inputs, try the middle steps with less help, then explain your trace. Check this claim before group discussion: The worked result establishes every future case

Explain the picture

Did the added cases demonstrate better answering? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Map user tasks and risks, author ten labeled cases, peer-audit missing coverage, then freeze expected behaviors.

Map user tasks and risks, author ten labeled cases, peer-audit missing coverage, then freeze expected behaviors.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY
1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Maintain a balanced eval set covering utility, grounding, format, privacy, and safety.

THIS WEEK’S CHECKPOINT Maintain a balanced eval set covering utility, grounding, format, privacy, and safety.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Author ten tests with expected behavior.

CHALLENGE Author ten tests with expected behavior.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 26 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Ten evaluation questions are easy definitions; none lacks a source.

The question

What is untested?

Worked explanation

Absent evidence, ambiguity and boundary handling. Add predefined fallback cases and keep final cases separate from tuning.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Empty evaluation categories reveal gaps

Evidence or stage · Value or result

Evidence or stage: Easy definitions

Value or result: 10

Evidence or stage: Absent evidence

Value or result: 0

Evidence or stage: Ambiguity

Value or result: 0

Evidence or stage: Boundary handling

Value or result: 0

Ten evaluation questions are easy definitions; none lacks a source.

What is untested?

Absent evidence, ambiguity and boundary handling. Add predefined fallback cases and keep final cases separate from tuning.

14. Try a new case independently

Independent case: Changed independent case

Approved notes: Robotics Tuesday Room2; Art Wednesday Room5. Requests: “Where does the club meet?” and “Who leads drama?” An unversioned added note says Robotics Room3. Design checks and safe responses.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 26 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

You improve a prompt after reading every test question. Can those questions still provide an untouched final evaluation? Propose a repair.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Maintain a balanced eval set covering utility, grounding, format, privacy, and safety.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A test set contains ten easy questions with direct answers in the notes. Add three different kinds of cases that probe fallback behaviour and define expected outcomes.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.